

Foundations of Neural Networks: McCulloch-Pitts, Perceptron & Gradient Descent

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Biological Motivation

- **Dendrite**: receives signals from other neurons
- **Synapse**: point of connection to other neurons
- **Soma**: processes information
- **Axon**: transmits the output of this neuron

Key Fact

The brain is a massively parallel interconnected network of $\approx 10^{11}$ neurons, arranged in a hierarchy. Each neuron may perform a certain role / respond to a certain stimulus.

McCulloch-Pitts (MP) Neuron (1943)

McCulloch (Neuroscientist) & Pitts (Logician) proposed a highly simplified computational model.

- Inputs $x_1, x_2, \dots, x_n \in \{0, 1\}$
- g aggregates: $g(x) = \sum_{i=1}^n x_i$
- f takes a binary decision:

$$y = f(g(x)) = \begin{cases} 1 & g(x) \geq \theta \\ 0 & g(x) < \theta \end{cases}$$

- θ = **thresholding parameter** $\Rightarrow$
Threshold Logic

Inhibitory Input Rule

If any x_i is inhibitory, $y = 0$ regardless of threshold.

Limitation

Only *linearly separable* Boolean functions. XOR cannot be represented!

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From MP Neuron to Perceptron

Four key advances (Rosenblatt):

- 1 **Weights** w_i on inputs (inputs are no longer equal)
- 2 **Weight learning mechanism** from data
- 3 **Real-valued inputs** (not limited to Boolean)
- 4 Refined & carefully analyzed by Minsky & Papert (1969)

$$y = \begin{cases} 1 & \sum_i w_i x_i \geq \theta \\ 0 & \sum_i w_i x_i < \theta \end{cases}$$

Setting $x_0 = 1$, $w_0 = -\theta$ (**bias node**):

$$y = \begin{cases} 1 & \mathbf{w}^T \mathbf{x} \geq 0 \\ 0 & \mathbf{w}^T \mathbf{x} < 0 \end{cases}$$

Perceptron Learning Algorithm

Algorithm 1 Perceptron Learning (normalized)

Input: P = inputs with label 1; N = inputs with label 0

Output: Weight vector $\mathbf{w}$

- 1: $P' \leftarrow P \cup N^-$ (where N^- contains negatives of all points in N)
 - 2: Initialize $\mathbf{w}$ randomly
 - 3: **while** not converged **do**
 - 4: Pick random $\mathbf{p} \in P'$; normalize: $\mathbf{p} \leftarrow \mathbf{p} / \|\mathbf{p}\|$
 - 5: **if** $\mathbf{w}^T \mathbf{p} < 0$ **then**
 - 6: $\mathbf{w} \leftarrow \mathbf{w} + \mathbf{p}$
 - 7: **end if**
 - 8: **end while**
-

- $\mathbf{w}^T \mathbf{x} = 0$ defines the decision hyperplane; $\mathbf{w}$ is orthogonal to it.
- Each correction rotates $\mathbf{w}$ closer (in angle) to the misclassified point.

Convergence Proof — Setup & Statement

Theorem 1 (Perceptron Convergence)

*If P and N are finite and linearly separable, the perceptron learning algorithm makes at most $k_{\max}$ corrections and **converges** in finite steps.*

Setup:

- Let $\mathbf{w}^*$ be the (normalized) optimal solution: $\|\mathbf{w}^*\| = 1$.
- Since data is linearly separable, $\exists \delta > 0$ such that $\mathbf{w}^{*T} \mathbf{p} \geq \delta$ for all normalized $\mathbf{p} \in P'$.
- Let β_k = angle between $\mathbf{w}^*$ and $\mathbf{w}_k$ after k corrections.
- Track:

$$\cos \beta_k = \frac{\mathbf{w}^* \cdot \mathbf{w}_k}{\|\mathbf{w}^*\| \|\mathbf{w}_k\|} = \frac{\mathbf{w}^* \cdot \mathbf{w}_k}{\|\mathbf{w}_k\|}$$

Convergence Proof — Numerator Grows Linearly

At step t : correction is $\mathbf{w}_{t+1} = \mathbf{w}_t + \mathbf{p}_t$ (triggered because $\mathbf{w}_t^T \mathbf{p}_t < 0$).

Effect on numerator $\mathbf{w}^* \cdot \mathbf{w}_{t+1}$:

$$\begin{aligned}\mathbf{w}^* \cdot \mathbf{w}_{t+1} &= \mathbf{w}^* \cdot (\mathbf{w}_t + \mathbf{p}_t) \\ &= \mathbf{w}^* \cdot \mathbf{w}_t + \underbrace{\mathbf{w}^* \cdot \mathbf{p}_t}_{\geq \delta} \\ &\geq \mathbf{w}^* \cdot \mathbf{w}_t + \delta\end{aligned}$$

By induction over k corrections (starting from $\mathbf{w}_0$):

$$\boxed{\mathbf{w}^* \cdot \mathbf{w}_k \geq \mathbf{w}^* \cdot \mathbf{w}_0 + k\delta}$$

The numerator grows **at least linearly** in k .

Convergence Proof — Denominator Grows Sub-Linearly

Effect on $\|\mathbf{w}_{t+1}\|^2$:

$$\begin{aligned}\|\mathbf{w}_{t+1}\|^2 &= \|\mathbf{w}_t + \mathbf{p}_t\|^2 \\ &= \|\mathbf{w}_t\|^2 + \underbrace{2\mathbf{w}_t^T \mathbf{p}_t}_{< 0} + \underbrace{\|\mathbf{p}_t\|^2}_{= 1 \text{ (normalized)}} \\ &\leq \|\mathbf{w}_t\|^2 + 1\end{aligned}$$

By induction after k corrections:

$$\boxed{\|\mathbf{w}_k\|^2 \leq \|\mathbf{w}_0\|^2 + k}$$

Observation

Denominator $\|\mathbf{w}_k\| \leq \sqrt{\|\mathbf{w}_0\|^2 + k}$, which grows only as $\sqrt{k}$ — **sub-linearly** in k .

Convergence Proof — Conclusion

Combining both bounds:

$$\cos \beta_k = \frac{\mathbf{w}^* \cdot \mathbf{w}_k}{\|\mathbf{w}^*\| \|\mathbf{w}_k\|} \geq \frac{\mathbf{w}^* \cdot \mathbf{w}_0 + k\delta}{\sqrt{\|\mathbf{w}_0\|^2 + k}} \xrightarrow{k \rightarrow \infty} \infty$$

This grows proportionally to $\sqrt{k} \cdot \delta$.

Contradiction!

$\cos \beta_k \leq 1$ by definition of cosine — it can never exceed 1.

Therefore

k must be **bounded above** by some finite $k_{\max}$:

$$k \leq k_{\max} = \frac{(\mathbf{w}^* \cdot \mathbf{w}_0)^2 + \|\mathbf{w}_0\|^2 \delta^2}{\delta^2} \quad (\text{approx.})$$

$\Rightarrow$ Finite corrections $\Rightarrow$ Algorithm **converges**.



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Sigmoid Neuron

Perceptron: hard threshold, binary output.

Sigmoid Neuron: smooth, real-valued output in $(0, 1)$:

$$y = \sigma(\mathbf{w}^T \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x}}}$$

- $\mathbf{w}^T \mathbf{x} \rightarrow +\infty \Rightarrow y \rightarrow 1$
- $\mathbf{w}^T \mathbf{x} \rightarrow -\infty \Rightarrow y \rightarrow 0$
- $\mathbf{w}^T \mathbf{x} = 0 \Rightarrow y = \frac{1}{2}$

Key Property

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$

Output as *probability* enables gradient-based learning — no sharp jump around threshold $(-w_0)$.

Loss Function

$$\mathcal{L}(\mathbf{w}, b) = \frac{1}{N} \sum_{i=1}^N (y_i - \sigma(x_i))^2$$

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Taylor Series — The Foundation

Scalar form:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \dots$$

Rewriting with $y = x + h$:

$$f(y) = f(x) + (y-x)f'(x) + \frac{(y-x)^2}{2}f''(x) + \dots$$

Multivariate form (gradient ∇f , Hessian H):

$$f(y) = f(x) + (y-x)^T \nabla f(x) + \frac{1}{2}(y-x)^T H(x)(y-x) + \dots$$

Setting $\mathbf{w}_{k+1} = \mathbf{w}_k + \mathbf{s}$:

$$f(\mathbf{w}_{k+1}) = f(\mathbf{w}_k) + \mathbf{s}^T \nabla f(\mathbf{w}_k) + \frac{1}{2}\mathbf{s}^T H\mathbf{s} + \dots$$

(Here ∇f and H are evaluated at $\mathbf{w}_k$.)

GD via Taylor Series

Apply to loss $J(\mathbf{w})$ with step $s = \mathbf{w}_{k+1} - \mathbf{w}_k$:

$$J(\mathbf{w}_{k+1}) = J(\mathbf{w}_k) + (\mathbf{w}_{k+1} - \mathbf{w}_k)^T \nabla J(\mathbf{w}_k) + \frac{1}{2} (\mathbf{w}_{k+1} - \mathbf{w}_k)^T H(\mathbf{w}_{k+1} - \mathbf{w}_k) + \dots$$

Compactly:

$$J(\mathbf{w}_{k+1}) = J(\mathbf{w}_k) + s^T \nabla J + \frac{1}{2} s^T H s + \dots$$

Gradient Descent Update

$$s = -\eta \nabla J(\mathbf{w}_k) \iff \mathbf{w}_{k+1} = \mathbf{w}_k - \eta \nabla J(\mathbf{w}_k)$$

Why Does GD Work? — Non-Increasing Updates

Use the **first-order** Taylor approximation and substitute $s = -\eta \nabla J$:

$$\begin{aligned} J(\mathbf{w}_{k+1}) &\approx J(\mathbf{w}_k) + s^T \nabla J(\mathbf{w}_k) \\ &= J(\mathbf{w}_k) + (-\eta \nabla J)^T \nabla J \\ &= J(\mathbf{w}_k) - \underbrace{\eta \|\nabla J(\mathbf{w}_k)\|^2}_{\geq 0} \end{aligned}$$

Conclusion

For any $\eta > 0$:

$$J(\mathbf{w}_{k+1}) \leq J(\mathbf{w}_k) \quad \Rightarrow \quad \textbf{Non-increasing at every step.}$$

When $\nabla J = \mathbf{0}$: no change in objective $\Rightarrow$ iterations stop at a stationary point.

Why Steepest Descent? — Directional Derivative

From the first-order Taylor expansion:

$$J(\mathbf{x} + \Delta\mathbf{x}) \approx J(\mathbf{x}) + \Delta\mathbf{x}^T \nabla J(\mathbf{x})$$

The **directional derivative** along unit vector $\mathbf{u}$ is $\mathbf{u}^T \nabla J$.

Best direction to decrease J ?

$$\min_{\mathbf{u}, \|\mathbf{u}\|=1} \mathbf{u}^T \nabla J = \min_{\mathbf{u}, \|\mathbf{u}\|=1} \|\mathbf{u}\| \|\nabla J\| \cos \theta$$

Minimized when $\cos \theta = -1$, i.e., $\mathbf{u}$ points **opposite** to ∇J .

Steepest Descent

$$\mathbf{x}_{\text{new}} = \mathbf{x}_{\text{old}} - \eta \nabla J$$

Moving opposite to the gradient gives the **steepest decrease** in J .

Is There an Optimal Learning Rate?

Using the **second-order** Taylor expansion with $s = -\eta \nabla J$:

$$J(\mathbf{w}_{k+1}) = J(\mathbf{w}_k) - \eta \nabla J^T \nabla J + \frac{\eta^2}{2} \nabla J^T H \nabla J$$

Differentiate w.r.t. η and set to zero:

$$\frac{d}{d\eta} J(\mathbf{w}_{k+1}) = -\|\nabla J\|^2 + \eta \nabla J^T H \nabla J = 0$$

$$\eta^* = \frac{\|\nabla J\|^2}{\nabla J^T H \nabla J}$$

Note

η^* changes at every iteration k since ∇J and H both depend on $\mathbf{w}_k$. Computing H at every step is expensive — motivates second-order methods.

Gradient, Hessian & Jacobian — Recap

Gradient

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Scalar output.
Steepest ascent
direction.

Hessian

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

Curvature matrix.
Tells *how far* to trust
gradient.

Jacobian

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

$$J_{ij} = \frac{\partial f_i}{\partial x_j}$$

Generalizes gradient to
vector-valued
functions.

Second-Order Methods (Newton's Method)

Idea: find the step s that minimizes the second-order Taylor expansion.

Differentiate $J(\mathbf{w}_k) + s^T \nabla J + \frac{1}{2} s^T H s$ w.r.t. s and set to zero:

$$\nabla J + Hs = 0 \implies s = -H^{-1} \nabla J$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k - H^{-1} \nabla J(\mathbf{w}_k)$$

Advantages

- Often **quadratic convergence**
- Second-order approx. more accurate
- Gradient gives direction; Hessian gives magnitude

Disadvantages

- H is $d \times d$ — huge memory
- H^{-1} costs $O(d^3)$
- Negative eigenvalues (saddle points) cause ascent steps
- Fix: regularize as $(H + \alpha I)^{-1} \nabla J$

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Neural Network Foundations

- **MP Neuron**: threshold logic on Boolean inputs; only linearly separable functions
- **Perceptron**: weighted inputs + bias + learning; converges if data is linearly separable (proved via $\cos \beta$)
- **Sigmoid**: smooth probabilistic output enabling gradient flow

Optimization Foundations

- **Taylor Series**: base for all GD analysis
- **GD**: non-increasing updates; steepest descent is optimal 1st-order direction
- **Optimal** η : from 2nd-order Taylor
- **Newton's**: faster but costly H^{-1} ; regularize for saddle points

“Gradient tells direction; Hessian tells how much to trust it.”

Thank You

Questions?